

ENHANCING SINGLE-CELL RNA-SEQ ANALYSIS THROUGH THE INTEGRATION OF UNACCOUNTED INTERGENIC REGIONS

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1. LIST OF ABBREVIATIONS

cDNA	complementary DNA
CPM	counts per million
GRN	gene regulatory network
NGS	next generation sequencing
PBMC	Peripheral Blood Mononuclear Cells
RT	reverse transcription
scRNA-seq	single cell RNA sequencing
UMI	unique molecular identifier

2. INTRODUCTION

Single-cell RNA sequencing (scRNA-seq) is becoming an increasingly popular tool for analyzing cellular systems. This technology enables whole transcriptome sequencing of thousands of cells in a single experiment, generating a vast amount of data. The typical workflow for analyzing such data involves mapping reads to a known genome reference and constructing cell-gene matrices, i.e., matrices in which rows correspond to cells, columns correspond to genes, and each entry indicates the expression level of a gene in a specific cell. Such matrices are used in downstream analyses. However, there are some reads that map to the genome but remain unassigned to any known gene. As a result, such reads are not used in downstream analysis.

Since these unassigned reads can constitute a significant fraction of the total read count (often up to 30%), understanding their origin could help improve scRNA-seq technologies and analysis.

There are two potential sources of unassigned reads: either they are sequencing artifacts, or they originate from real transcripts that remain unassigned due to issues related to transcriptomic references.

The two main challenges with transcriptomic references are:

1. Incomplete transcriptome annotations – even though great efforts have been made to annotate all genes, it is very likely that not all genes are annotated, and many remain to be found. These unannotated genes are missed in the typical scRNA-seq analysis.
2. Complexity and overlaps in transcriptomic features – the genomes of humans and many other species are highly complex, with many overlapping features. In such cases, mapping algorithms assign multiple features for a given read, and such reads are usually discarded from downstream analysis.

This project focuses on unassigned reads in scRNA-seq datasets, aiming to uncover their origins and enhance scRNA-seq data analysis by incorporating biologically meaningful data that is typically disregarded.

3. AIM AND TASKS

Aim This project aims to determine whether including unassigned reads can improve scRNA-seq analysis.

Tasks

1. Identify genomic regions containing unassigned reads, and classify them as either intersecting with genes or intergenic.
2. For intergenic regions, collect evidence to confirm whether they represent technical noise or unannotated genes.
3. For intersecting regions, attempt to resolve problems related to the transcriptomic reference that lead to reads being unassigned.

4. LITERATURE REVIEW

4.1 Introduction to single cell transcriptomics

Cells are the fundamental units of life, forming the basis of all living organisms. One of the major goals of biology is to understand cellular systems and the processes occurring within cells. Since the discovery of the DNA structure in 1953 and the development of the conceptual framework for genetic information transfer (Watson and Crick 1953), scientists have made significant efforts to sequence the genomes of various organisms. This led to the development of the first sequencing methods, such as Sanger sequencing in 1975 (Sanger, Nicklen, and Coulson 1977), which laid the foundation for next-generation sequencing (NGS) technologies in use today, such as the widely used Illumina platform (Heather and Chain 2016). Current sequencing methods allow us to obtain the complete genetic sequence of any organism. However, the genome alone cannot explain the full diversity of cells in multicellular organisms, as all cells share the same genome but exhibit significant variation in shape, size, and function.

RNA sequencing (RNA-seq), on the other hand, enables the measurement of gene expression within cells, providing valuable insights into cellular processes. RNA-seq methods largely follow DNA sequencing protocols, with the addition of a step in which complementary DNA (cDNA) is synthesized from RNA (Heumos et al. 2023). The first RNA-seq methods were developed for bulk sequencing, where RNA from entire cell populations is sequenced, providing an average gene expression profile across the population. Although bulk RNA-seq has provided valuable insights into the dynamics of cellular processes (such as changes in disease states in response to therapeutics, detection of gene isoforms, gene fusions, and various other properties of target cells (Heumos et al. 2023)), this approach masks non-dominant processes and cell-to-cell variability through averaging. This limitation was addressed by the introduction of single-cell RNA sequencing (scRNA-seq) methods, which allow for the generation of transcriptomic profiles from individual cells, providing high-resolution insights into cellular systems.

Current scRNA-seq methods enable the generation of transcriptomic profiles from thousands of cells at unprecedented resolution in a single experiment. These data can be used for constructing cellular atlases (Rozenblatt-Rosen et al. 2017), understanding disease mechanisms (Z. Zhang, Chen, and X. Peng 2024), exploring cell differentiation and developmental processes (Skinner, Asad, and Haque 2024), among other applications.

4.2 scRNA-seq data generation and analysis

4.2.1 Overview of scRNA-seq protocols

The generation of scRNA-seq data is a complex, multi-step process that varies across different protocols. These protocols can be grouped based on several criteria, such as the type of RNA capture (e.g., 3' end, 5' end, or full-length) or the method of cell isolation (e.g., droplet-based methods such as inDrops (Klein et al. 2015), Drop-seq (Macosko et al. 2015), and Chromium by 10X Genomics (Zheng et al. 2017), or plate-based methods such as CEL-Seq2 (Hashimshony et al. 2016) and Smart-seq2 (Picelli et al. 2013)).

In this project, datasets generated using droplet-based 3' end sequencing methods were analyzed. Therefore, these methods will be the focus of the following literature review.

3' End Sequencing and Polyadenylation 3' end sequencing captures RNA molecules using primers complementary to poly(A) tails. Polyadenylation at the 3' end is a post-transcriptional modification in which non-templated adenosines are added to the 3' end of mRNA molecules. Although the poly(A) tail is present in almost all mRNAs, its length varies and can influence mRNA fate, stability, and translation efficiency (Brouze et al. 2022).

3' end sequencing protocols exploit this feature to selectively capture RNA molecules, enabling high-throughput and cost-effective sequencing. However, some RNA molecules lack poly(A) tails and are therefore not captured by these methods, such as replication-dependent histone mRNAs (Brouze et al. 2022).

Common Steps in scRNA-seq Protocols Despite differences between scRNA-seq protocols, they share key steps: single-cell isolation, library preparation, and sequencing (Andrews and Hemberg 2018). In droplet-based methods, single cells are encapsulated in individual droplets containing hydrogel-attached barcoding primers and a lysis mix (an example of a droplet generation device is shown in Figure 4.1). Primers used in these protocols typically share a common structure, including:

- **Cell barcodes:** Unique sequences that identify the cell from which a particular read originates (since all droplets are pooled and sequenced together).
- **Unique molecular identifiers (UMIs):** Short sequences used to quantify the original number of RNA molecules, helping to eliminate amplification bias.
- **PCR handles:** Sequences that facilitate amplification.
- **Poly-T sequences:** In 3' end sequencing methods are used to selectively capture polyadenylated RNAs (X. Zhang et al. 2019).

An example of primer design is shown in Figure 4.2. Once cells are encapsulated in droplets, lysis occurs, releasing RNA, which is then captured by the primers.

Depending on the method, reverse transcription may occur within the droplets (as in inDrops and 10X Genomics) or in bulk (as in Drop-seq). Subsequent steps typically include RNA fragmentation, PCR amplification, and next-generation sequencing (NGS).

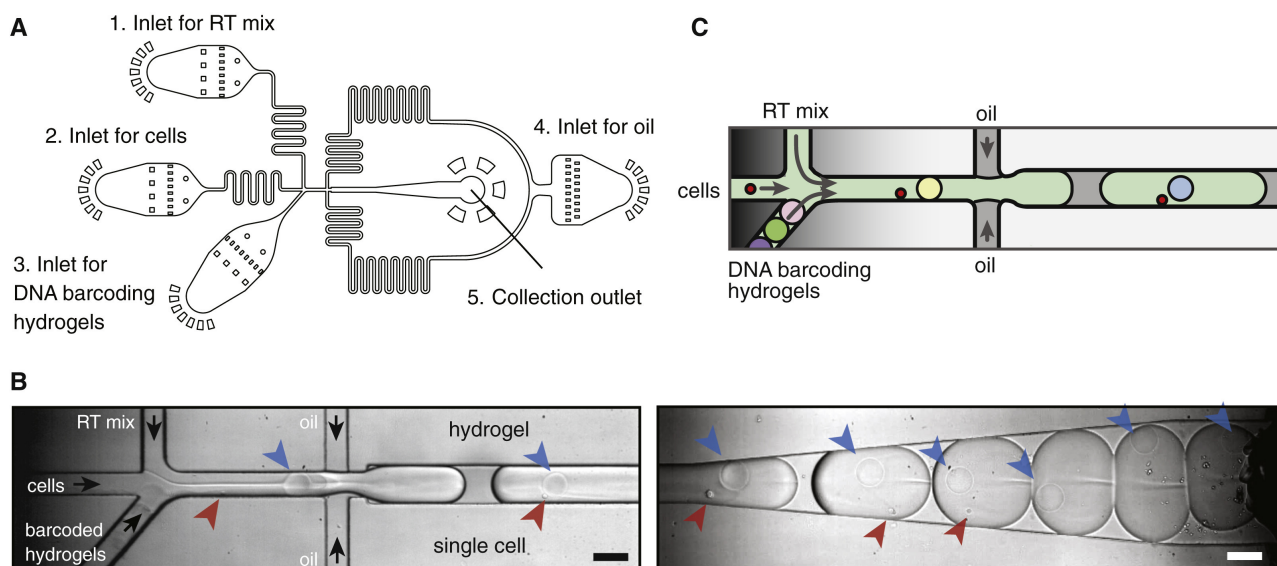


Figure 4.1: An example of microfluidics device for droplet generation (inDrops). A) Schematic view of the device. B) Snapshots of droplet generation and collection. C) Scheme of droplet generation. Figure taken from Klein et al. (2015).

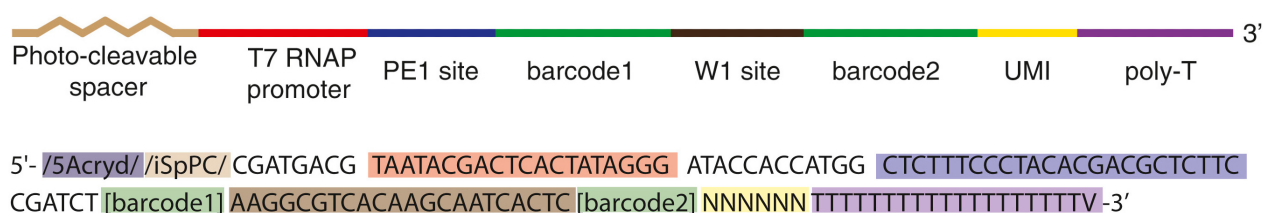


Figure 4.2: Primer design in the inDrops protocol.. The image above shows schematic view, while below is given example with sequences. The structure of primers varies between the protocols, the main parts (UMI site, barcodes, poly-T's) are found in most of them. Here also present are promoter region for RNA polymerase (red), sequencing primer (blue), synthesis adaptor (dark brown). Figure taken from Klein et al. (2015).

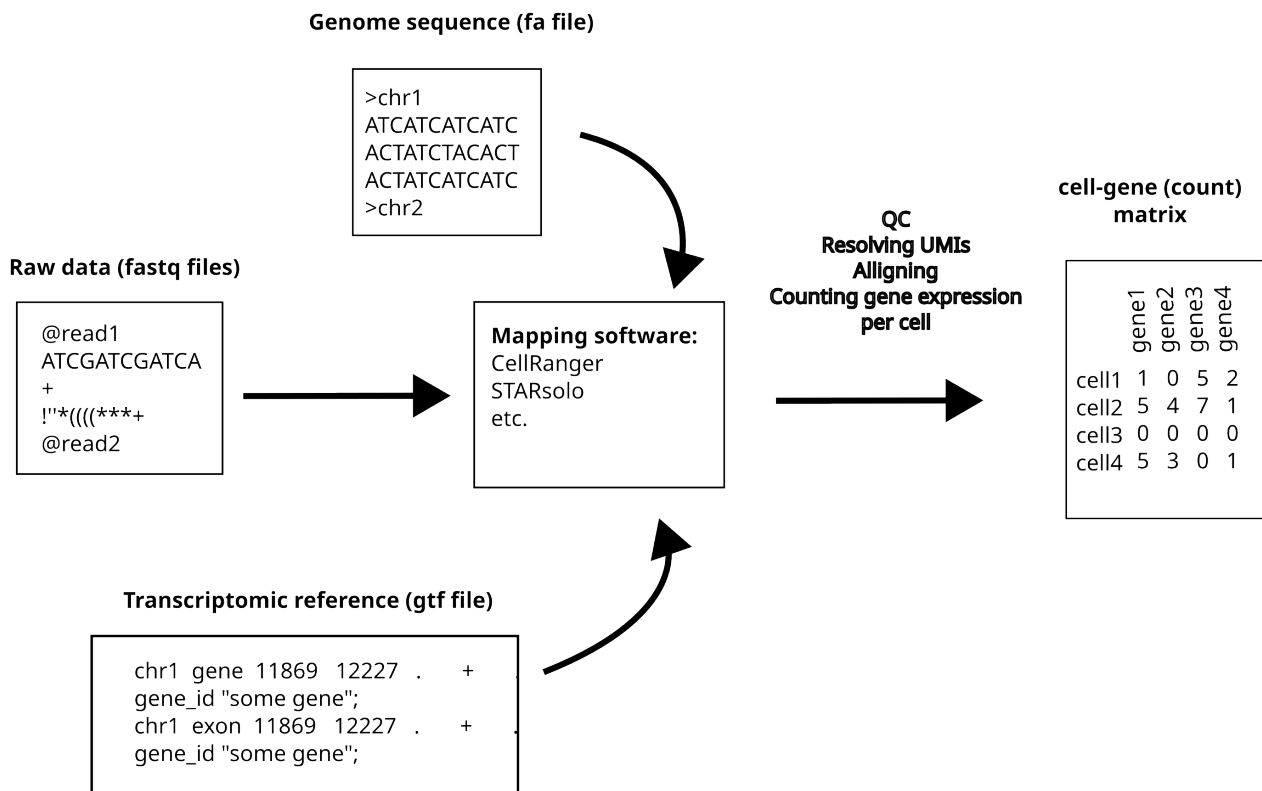


Figure 4.3: Inputs and outputs of mapping software.

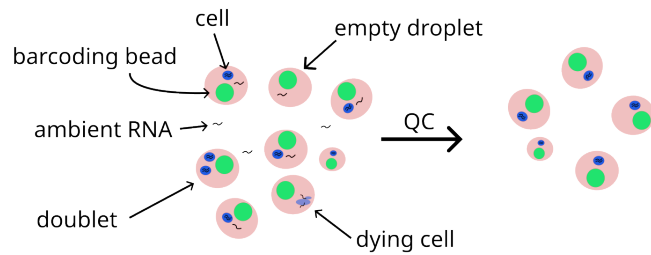
4.2.2 scRNA-seq data analysis

Raw data processing The output from NGS machines is typically converted to FASTQ files, containing recorded sequences, as well as (depending on method) barcode and UMI sequences, and quality scores. The subsequent processing steps include quality control of FASTQ file (based on quality scores), filtering duplicate reads (using UMIs), mapping reads to the genome sequence, assigning the reads to the genes, and finally, counting gene expression per cell (barcode) (Heumos et al. 2023) (see figure 4.3). Usually, all these steps are performed with a single piece of dedicated software, such as STARsolo (Kaminow, Yunusov, and Dobin 2021), CellRanger (Zheng et al. 2017) or other. It should be noted, that there are variations in the pipeline described above, depending on many experiment-related (e.g., whether the genome sequence or transcriptome of the study organism is known), or method-related (e.g., whether UMIs are used in the protocol) factors. The typical result of such processing is cell-gene matrix (i.e., a matrix where rows represent cells, columns represent genes, and each entry indicates the number of captured RNAs for a given gene in a specific cell).

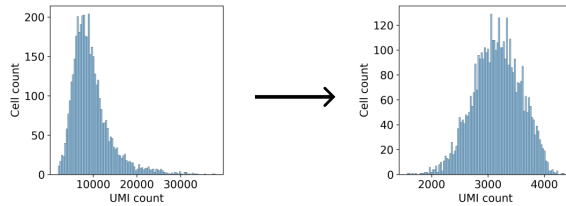
Cell-gene matrix processing The next steps in the analysis of the scRNA-seq data involve following steps (see Figure 4.4 for graphical summary):

- **Quality control** The quality of individual cells (represented as all reads harboring a unique barcode) can be evaluated based on several factors, such as mitochondrial gene content (apoptotic cells tend to have a higher proportion of mitochondrial transcripts (Heumos et al. 2023)) or total number of captured genes (very low numbers can be produced by ambient RNA entering empty

Quality control



Normalization



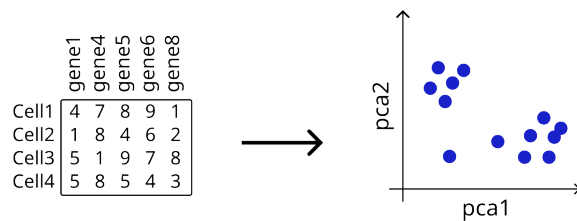
Feature selection



	gene1	gene2	gene3	gene4	gene5	gene6	gene7	gene8
Cell1	4	5	0	7	8	9	0	1
Cell2	1	5	0	8	4	6	0	2
Cell3	5	5	1	1	9	7	0	8
Cell4	5	5	0	8	5	4	0	3

	gene1	gene4	gene5	gene6	gene8
Cell1	4	7	8	9	1
Cell2	1	8	4	6	2
Cell3	5	1	9	7	8
Cell4	5	8	5	4	3

Dimensionality reduction



Clustering and other analyses

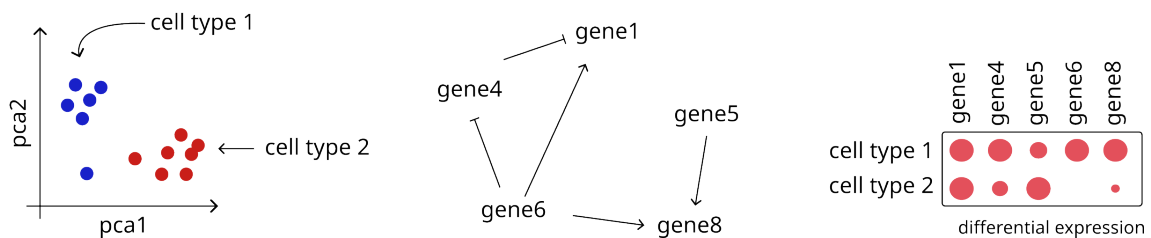


Figure 4.4: The typical data analysis pipeline. Firstly data is cleaned and normalized, then (to reduce dimensionality) non-informative genes are filtered out and dimensionality reduction methods are applied. Final steps depends on the focus of study, but often include cell clustering, annotation, differential gene expression analysis or gene regulatory network inference.

droplets). In some cases, two cells can end up in one droplet, producing a count matrix row that reflects the combined gene expression of both cells. Such matrix entries (doublets) can be filtered by using specialized software such as Scrublet (Wolock, Lopez, and Klein 2019) or scDblFinder (Germain et al. 2022). Another source of noise in scRNA-seq data is ambient RNA, which consists of RNA that escapes individual droplets and spreads into the medium or other droplets, leading to background noise. Even though the amount of such RNA is not high (in good quality datasets it can be around 2% (Young and Behjati 2020)), removing these RNAs from the count matrix can improve data quality. This can be achieved by identifying the background noise profile from empty droplets and adjusting the count matrix accordingly. There are dedicated softwares, such as SoupX (Young and Behjati 2020), decontX (Yang et al. 2020), CellBender (Fleming et al. 2023) and others.

- **Normalization** The next step in preprocessing pipeline is normalization. The goal of normalization is to transform the data so that the variation in gene expression levels is comparable, making subsequent analysis more efficient (Ahlmann-Eltze and Huber 2023). Normalization can also help eliminate biases, such as differences in sequencing depth when combining data from multiple samples (Lingen, Suarez-Diez, and Saccenti 2024). There are numerous normalization methods, based on different approaches (e.g., delta-method-based, residual-based, latent gene expression-based, count-based (Ahlmann-Eltze and Huber 2023)). Thus, selecting a normalization method should be done carefully, depending on the experimental design. General recommendations for normalization suggest comparing several methods, and if the results are similar, opting for the simpler method (Lingen, Suarez-Diez, and Saccenti 2024). Sophisticated methods do not necessarily show better results, and a recent benchmarking study by Ahlmann-Eltze and Huber (2023) has shown that simpler method (particularly the logarithm normalization, where each element y of the count matrix is transformed by formula $y_{transformed} = \log(y + 1)$) performs as well or better than more advanced methods.
- **Filtering genes** Once the data is normalized and cleaned, one can filter out non-informative genes. Initially, count matrices contain all the genes that are present in the transcriptome. However, not all of them are expressed in the sequenced data, or are expressed in negligible numbers (Heumos et al. 2023). Therefore, it is common practice to filter such genes (e.g., genes that are expressed in less than three cells). Moreover, some genes might be expressed in all the cells more or less evenly (housekeeping genes), which do not provide useful information that could be useful in, for instance, grouping cells or determining cell types. Therefore, in many applications, it is beneficial to leave only those genes, that are highly variable between cells. In such way, the dimensionality of the count matrix is greatly reduced without losing significant information. Additionally, genes that are outside the scope of the specific study can also be filtered out.
- **Dimensionality reduction** Even after filtering and selecting only highly variable genes, several thousand genes usually remain. It is not feasible to visualize (and hard to interpret in general) data of such high dimensionality, therefore, dimensionality reduction is essential step of subsequent analysis. The idea of dimensionality reduction is simple: to reduce the dimensions of the data losing as little information as possible. There are a number of dimensionality reduction

methods based on different mathematical concepts, but the most widely used today include t-SNE (Hinton and Roweis 2002), UMAP (McInnes, Healy, and Melville 2018) and principal component analysis (PCA). Although the use of these algorithms are supported by some benchmarking studies (in the study of Xiang et al. (2021), t-SNE was showed best performance, while UMAP showed the highest stability), other benchmarking studies report different findings. The study of Koch et al. (2021) suggested that such overlooked methods as latent Dirichlet allocation (LDA) and PHATE are among the best performing (metrics for comparison included clustering accuracy, structure preservation, efficiency and others). Meanwhile Sun et al. (2019) provided guidelines for choosing dimensionality reduction method depending on downstream analysis tasks, and in their results UMAP and tSNE were not on the top choices. Thus, while UMAP remains the gold standard method in the field (usually preceded by PCA for initial dimensionality reduction), it is worth considering alternative methods as well.

- **Clustering and other analyses** One of the most common tasks of scRNA-seq data analysis is to identify and classify cell populations (Andrews and Hemberg 2018). This task requires to assign cells to different groups (clusters), such that cells in the same clusters are similar and distinct from cells in other clusters. There is a great variety of clustering algorithms available, including k-means, hierarchical and consensus clustering (L. Peng et al. 2020). Benchmarking studies suggest that "no individual scRNA-seq clustering algorithm can capture true clusters and achieve optimal performance in all situations" (L. Peng et al. 2020).

Clustering is usually followed by cell annotation (i.e., assigning cell type to the identified clusters), which is done by finding cell type specific markers or using automatic (machine learning) tools such as CellTypist (Domínguez Conde et al. 2022). The subsequent steps in the analysis depend on the focus of the particular study and can include differential gene expression (dge) analysis, investigation of the dynamics of cellular systems (RNA velocity, pseudotime), inferring gene regulatory networks (GRNs), and more.

The described analysis pipeline enables insights into multicellular systems. All downstream analyses rely heavily on the initial processing steps, particularly read mapping, which depends on the transcriptomic reference. In the next sections, I will shortly introduce genome biology and further expand on transcriptomic references.

4.3 Genome biology

An organism's biological features are largely encoded in its genome. The genome is typically defined as the complete set of genetic material (usually DNA) of an organism. DNA is a double-stranded helical molecule composed of four nucleotides: adenine (A), cytosine (C), guanine (G), and thymine (T). The strands are complementary: adenine pairs with thymine, and cytosine pairs with guanine (Alberts et al. 2022). DNA serves as a template for RNA transcription, which can then be translated into proteins or function directly as RNA (e.g., regulatory RNAs).

Genes are generally defined as genomic regions transcribed into functional RNAs. For some RNAs, parts of the transcript are removed in a process known as splicing. The discarded segments are called introns, while the retained, coding parts are called exons. It is worth noting that only a small portion

of the human genome is transcribed. For example, in the GENCODE v47 annotation, exons span only about 6.5% of the genome sequence.

Beyond clearly functional genomic elements (e.g., protein-coding genes), there are non-coding regions that are still functional, such as transcription promoter regions. Additionally, there are regions that are transcribed but whose function remains unknown or may be nonfunctional (e.g., transcribed pseudogenes).

Not all regions of the eukaryotic genome — especially the human genome — are unique. In humans, more than half of the genome consists of repetitive sequences (Liehr 2021). These were historically dismissed as 'junk', 'selfish', or 'parasitic' DNA. However, later studies have shown that repetitive sequences can serve structural roles or contribute to genomic stability through redundancy (Shapiro and Sternberg 2005). Many of these repeats originate from transposons — mobile genetic elements capable of copying or repositioning themselves within the genome. Although transposons were initially considered parasitic, growing evidence suggests they play a generative role in genome evolution and should not be dismissed as 'junk' DNA (Fedoroff 2012).

Transcription Transcription is carried out by RNA polymerases—enzymes that synthesize RNA molecules from a DNA template. This process can be divided into three stages:

- **Initiation** – RNA polymerase recognizes the promoter region at the start of a gene and binds to it. The enzyme then unwinds the DNA duplex, begins RNA synthesis, and escapes the promoter (Cramer 2019).
- **Elongation** – RNA polymerase moves along the DNA template, synthesizing the RNA molecule.
- **Termination** – Upon encountering a termination signal, the polymerase detaches from the DNA and releases the completed RNA molecule, marking the end of transcription (Lodish et al. 2021).

Each of these steps involves additional proteins and RNA molecules, collectively known as transcription factors. For example, RNA polymerases cannot independently recognize promoter sequences, instead, specific transcription factors are required to recruit the polymerase to the promoter (Cramer 2019).

Chromatin Chromatin is a dynamic complex of DNA and histone proteins that packages the eukaryotic genome within the nucleus. Its fundamental unit, the nucleosome, comprises approximately 147 base pairs of DNA wrapped around a histone octamer, enabling efficient genome compaction and regulation (Alberts et al. 2022). Chromatin exists in a spectrum of conformational states, from tightly packed heterochromatin (generally transcriptionally inactive) to more relaxed euchromatin (transcriptionally active). Accessible chromatin refers to regions where DNA is less tightly associated with histones, permitting interactions with transcription factors and other regulatory proteins. Although accessible chromatin constitutes only 2–3% of the total genome sequence, it accounts for over 90% of known transcription factor binding sites (Klemm, Shipony, and Greenleaf 2019). Techniques such as ATAC-seq and DNase-seq are widely used to map these regions, offering insights into gene regulation and cellular identity (Minnoye et al. 2021).

Non-coding RNAs As mentioned earlier, not all transcribed RNAs are translated into proteins. In fact, a substantial portion of the genome is transcribed into non-coding RNAs, which include a wide

range of functional RNA types. Among these, long non-coding RNAs (lncRNAs) have emerged as a major output of complex eukaryotic genomes (Mattick et al. 2023). These RNAs are typically longer than 200 nucleotides and are not translated into proteins, yet they participate in essential biological processes.

Non-coding RNAs can serve both housekeeping and regulatory roles. Housekeeping ncRNAs include rRNAs, tRNAs, and small nuclear RNAs, while regulatory ncRNAs—particularly lncRNAs — are involved in gene expression regulation, chromatin remodeling, RNA splicing, translation, and even phase separation in the nucleus. lncRNAs are often transcribed from intergenic regions, but they can also arise from introns, antisense strands, pseudogenes, and enhancers (Poliseno, Lanza, and Pandolfi 2024).

These RNAs tend to be expressed in a cell type-specific and developmentally regulated manner, contributing to lineage specification, tissue identity, and responses to environmental cues. Despite their relatively low abundance and rapid sequence evolution, lncRNAs often exhibit conserved structure-function features, such as modular domains and repeat elements that mediate interactions with chromatin, RNA, and proteins. Importantly, they are frequently associated with chromatin-modifying complexes and are increasingly recognized as integral to the spatial and temporal organization of gene regulation during development (Mattick et al. 2023).

Given their diversity and specificity, the biological importance of non-coding RNAs—especially lncRNAs — is increasingly appreciated, and many are now considered as genes with crucial regulatory roles in development, physiology, and disease.

4.4 Genome annotation

The genome annotation process typically refers to the identification and mapping of genes within a given genome sequence (Guigó 2023). While the definition itself is straightforward, the process is highly complex. This is evident from the fact that, even more than 20 years after the first human genome assembly, human genome annotations are continuously updated with new transcripts and are expected to evolve further (Mudge et al. 2024). Figure 4.5 illustrates the changes in GENCODE annotation over time.

It is important to note that many medical and scientific research efforts rely on an accurate human gene list. Examples include genome-wide association studies (GWAS), which attempt to link genomic variants to nearby genes; RNA-seq analysis; and exome sequencing projects that use capture kits targeting most known exons (Pertea et al. 2018).

Currently, the two most widely used genome annotations are GENCODE (Mudge et al. 2024) and RefSeq (O’Leary et al. 2015) (maintained by NCBI). Despite their status as mature genome references, they report different numbers of genes. For instance, RefSeq includes 20,078 protein-coding genes (NCBI 2025), whereas GENCODE contains 19,868 (Ensembl 2025). This discrepancy highlights the ongoing challenge of accurately annotating genomes.

Genome annotation process RNA sequencing (RNA-seq) is the primary tool used in genome annotation. Full-length RNA sequencing allows for the capture of RNA molecules, which, when aligned to a reference genome, help construct the genome annotation of a given species (Salzberg 2019).

However, RNA-seq has limitations, the most significant being its inability to capture all RNA molecules. This poses particular challenges for detecting rare transcripts, which may either be treated as noise or not captured at all (Salzberg 2019). Therefore, bioinformatics tools are necessary to complement RNA-seq data (Guigó 2023), and most modern genome annotation pipelines integrate computational methods alongside sequencing data.

Computational genome annotation methods can be broadly categorized into two types: comparative annotation, which leverages the fact that protein-coding sequences tend to be more evolutionarily conserved, and *ab initio* annotation, which uses known sequence biases to predict genes (Guigó 2023). Despite significant advancements, genome annotation remains imperfect.

While the number of protein-coding genes is reaching a consensus — major databases report around 19,000 to 20,000 protein-coding genes — the number of non-coding genes is expected to increase in the future (Amaral et al. 2023). This is largely due to the structured nature of protein-coding genes (e.g., open reading frames, codon biases that skew nucleotide distributions), which makes them easier to detect using computational tools (Guigó 2023). Additionally, protein-coding genes have historically received greater attention because of their direct links to phenotype, leading to more experimental validation.

Current challenges in human genome annotation Despite extensive efforts, a fully comprehensive human genome annotation has not yet been achieved. Several factors contribute to this challenge.

Firstly, the complexity of the human genome itself presents significant obstacles. Compared to the total genome length, the number of genes is relatively low, and their sequences are interrupted by introns (Salzberg 2019). Additionally, bulk RNA sequencing often has relatively low sequencing depth, making it difficult to detect rare transcripts or those expressed in a cell-type-specific manner (Guigó 2023).

In principle, full-length single-cell RNA sequencing (scRNA-seq) could help overcome some of these limitations. However, it still has inherent biases introduced during library preparation, including those related to RNA processing status, post-transcriptional modifications, transcript length, cellular localization, and structural features (Guigó 2023). Short-read scRNA-seq typically provides higher throughput (Heumos et al. 2023), enabling the detection of rare transcripts. However, it does not capture full transcript structures, limiting its utility to supporting evidence, such as validating computationally predicted genes or indicating transcriptional activity at specific genomic locations.

Beyond technical limitations, there are ontological challenges, such as defining what constitutes a gene. While genes are often regarded as well-defined, discrete entities, the reality is more complex. Coding and non-coding transcripts frequently overlap in intricate arrangements with unclear boundaries, suggesting that transcripts may not be discrete, countable units but instead form a transcriptional continuum (Salzberg 2019). Additionally, the classification systems used in genome annotations may be partially artificial. For example, some pseudogenes, generally considered non-functional copies of functional genes, are transcribed and may have biological functions (Pei et al. 2012). Similarly, the distinction between protein-coding and non-coding genes is debated, as many protein-coding loci generate both coding and non-coding transcripts, and numerous long non-coding RNAs (lncRNAs) contain potentially coding open reading frames (ORFs) (Salzberg 2019).

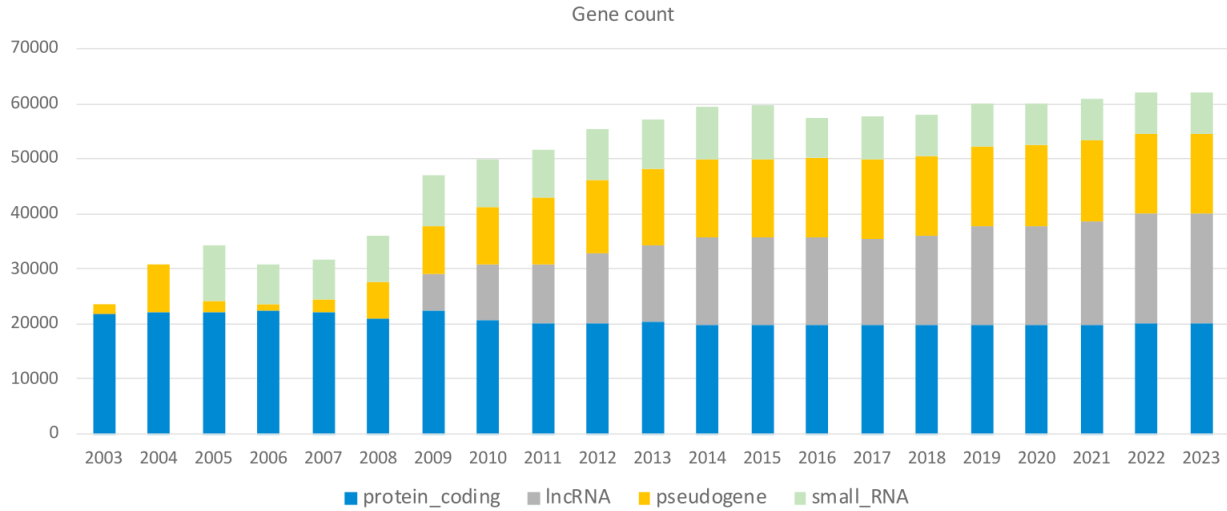


Figure 4.5: Changes in the number of annotated genes in GENCODE over time. Figure taken from paper by Guigó (2023)

Future perspectives of annotating genomes The size of eukaryotic genomes necessitates the use of automated methods for genome annotation. The accuracy of such methods depends directly on RNA capture technologies. If highly accurate and sensitive methods are developed, high-quality genome annotations can be expected (Salzberg 2019). However, at present, manual curation of human genome annotations remains essential due to the imperfections of both sequencing technologies and computational tools. Given the critical role of accurate gene annotation in both medical and scientific applications, continued improvements in annotation methods remain a priority.

4.5 Transcriptomic references for scRNA-seq

A well-annotated genome does not eliminate all challenges during read mapping. In simple terms, mapping algorithms determine the genomic location to which a sequence aligns, and if an annotated gene is present at that location, the read is assigned to that gene. However, there are cases where assigning a read to a gene is not straightforward, such as when a read maps to multiple locations or to a unique location where overlapping genes are annotated.

Multimappers There are two possible reasons for reads assigned to multiple genes: either they are mapped to multiple genome locations, or they are mapped to the genomic location containing overlapping features (genes). Such reads (commonly referred to as multimappers) are typically excluded from analysis (Almeida da Paz, Warger, and Taher 2024). However, as demonstrated by (Almeida da Paz, Warger, and Taher (2024)), this approach can introduce biases that affect downstream analyses. Currently, various methods exist to handle such cases (Deschamps-Francoeur, Simoneau, and Scott 2020), yet no entirely satisfactory solution has been established (Almeida da Paz, Warger, and Taher 2024).

The aforementioned methods primarily rely on computational strategies to resolve multimappers, but the issue can also be approached from a transcriptomic reference perspective. For example, in the case of CellRanger (the mapping software from 10x Genomics) the GENCODE reference is filtered

based on gene types (e.g., retaining protein-coding genes while filtering out pseudogenes) (10x Genomics 2025). While this filtering improves the number of reads included in downstream analyses, there is still room for improvement.

Pool et al. (2023) proposed three key steps to enhance transcriptomic references:

1. Including reads mapped to intronic sequences in the analysis.
2. Extending the 3' ends of certain genes.
3. Resolving overlaps between specific genes.

The first suggestion is not new in scRNA-seq research. Concepts such as RNA velocity, which rely on the ratio of spliced to unspliced RNA (La Manno et al. 2018), demonstrate that including intronic reads can provide valuable information. Moreover, most mapping tools (e.g., STARsolo, CellRanger) offer options to align reads either to exonic regions only or to entire genes. The second suggestion is based on the observation that scRNA-seq data often exhibits peaks of reads just beyond the 3' ends of genes. While the exact biological reasons remain unclear — possibly due to imprecise annotations — it is reasonable to associate these reads with the nearest genes. The third suggestion addresses gene overlaps. Reads originating from overlapping regions are often unassigned to any gene, yet in some cases, they are more likely to come from one gene rather than another. Overlapping gene resolution seeks to correct this by modifying the transcriptomic reference, either by shortening or removing certain genes.

Although Pool et al. (2023) proposed a tool to implement these strategies, it has limitations. Some of its aspects remain debatable (e.g., threshold choices), others seem unnecessary (such as handling exon-intron distinctions when most alignment tools already provide this option), and the process still requires significant manual effort. Thus, there is still a need for a more comprehensive tool for enhancing transcriptomic references — a need that will be addressed in this thesis.

Reads mapped to the intergenic regions Some reads map to the genome but not to any annotated gene. While these could simply be noise in scRNA-seq data, there is also the possibility that they contain biologically relevant information. As will be shown in the results section, scRNA-seq datasets indeed contain such reads, and they are not entirely noise.

5. METHODS

5.1 Data Acquisition

scRNAseq datasets We analyzed datasets from four different human tissues: brain, blood, lung, and eye. All samples, except for two Peripheral Blood Mononuclear Cell (PBMC) datasets, were generated using the 10x Genomics v3.1 protocol. The two exceptions were prepared using the inDrops2 protocol. All protocols capture 3' ends of RNA molecules and use short-read sequencing. The datasets are publicly available, with sources and additional details provided in Table 5.1.

sample name	tissue	cell count	donors	protocol	source
PBMC_10x	blood (PBMC)	5000	1	10x v3.1	10x genomics
PBMC_10x_2	blood (PBMC)	10000	1	10x v3.1	10x genomics
PBMC_10x_3	blood (PBMC)	10000	1	10x v3.1	10x genomics
PBMC_indrops	blood (PBMC)	2000	1	indrops2	Juzenas et al. 2025
PBMC_indrops_2	blood (PBMC)	9000	1	indrops2	Juzenas et al. 2025
brain	brain	6000	1	10x v3.1	Siletti et al. 2023
brain_2	brain	7000	1	10x v3.1	Siletti et al. 2023
eye	retina	10000	6	10x v3.1	Menon et al. 2019
eye_2	peripheral retina	2500	1	10x v3.1	Voigt et al. 2019
eye_3	peripheral retina	2500	1	10x v3.1	Voigt et al. 2019
lung_2	lung	5000	1	10x v3.1	Mould et al. 2021
lung_5	lung	5000	1	10x v3.1	Mould et al. 2021
lung_7	lung	4500	1	10x v3.1	Mould et al. 2021
lung_8	lung	5000	1	10x v3.1	Mould et al. 2021

Table 5.1: Datasets summary.

Genome and Transcriptomic references The human genome assembly GRCh38 was used in this project, downloaded from the [Ensembl website](#).

Four transcriptomic references were analyzed (see Table 5.2 for details). To ensure compatibility with the genome FASTA file, chromosome name prefixes ('chr') were removed from references that included them, as the genome file does not use these prefixes. This modification was performed using basic Linux command-line tools. Additionally, for the NCBI reference, chromosome names were converted to Ensembl-style notation using a custom Python script (e.g., 1, 2, 3 instead of NC_000001.11, NC_000002.12, NC_000003.12, etc.).

For references that do not have 'gene' entries, such entries were added (entries that span all the components of a particular gene).

reference name	source	version
10x	10x website	2024-A
GENCODE	Gencode website	47
RefSec (NCBI)	NCBI website	GCF_000001405.40
lnc	LNCipedia website	5.2

Table 5.2: References used in this project.

Gene Prediction Tracks and Conservation Scores Gene predictions (tracks 'AUGUSTUS', 'Geneid genes', 'Gescan genes', 'SGP genes', 'SIB genes') and genome conservation scores ('phast-Cons100way' track) were downloaded from [UCSC genome browser](#).

ATAC data The open chromatin regions (bed format) in PBMCs was downloaded from [10x website](#).

Oxford Nanopore data PBMC nanopore (full-length cDNA) data were downloaded from [10x genomics website](#).

Viral genomes Viral genomes for blasting unassigned sequences for BLAST searches were downloaded from [NCBI Refseq genomes database](#).

5.2 Computational Tools and Environment

All analyses were performed on a high-performance computing (HPC) cluster running a Linux environment. Software packages and tools were managed using Conda. The exact Conda environment specifications (YAML file) can be found on [GitHub repository](#). Besides tools managed by conda, also basic command line tools were used (e.g. *awk*, *wc*, *grep*, *sed*, *uniq*, *sort* and similar). The specific functions and parameters used are described in the following sections, where the general analysis pipeline will be explained.

5.3 Data processing pipeline

The pipeline was implemented using GNU MAKE and the full scripts are available on the [GitHub](#), here only general description of the workflow and used tools is provided. The general scheme of the pipeline (used both for extracting intergenic regions and enhancing transcriptomic reference) can be seen in Figure 5.1, while the description is given below:

1. Map reads with initial transcriptomic reference.
2. Take unassigned (and uniquely mapped) reads.
3. Split into intersecting and intergenic reads.
 - (a) For intersecting:
 - i. Resolve overlapping genes that have unassigned reads (if possible).

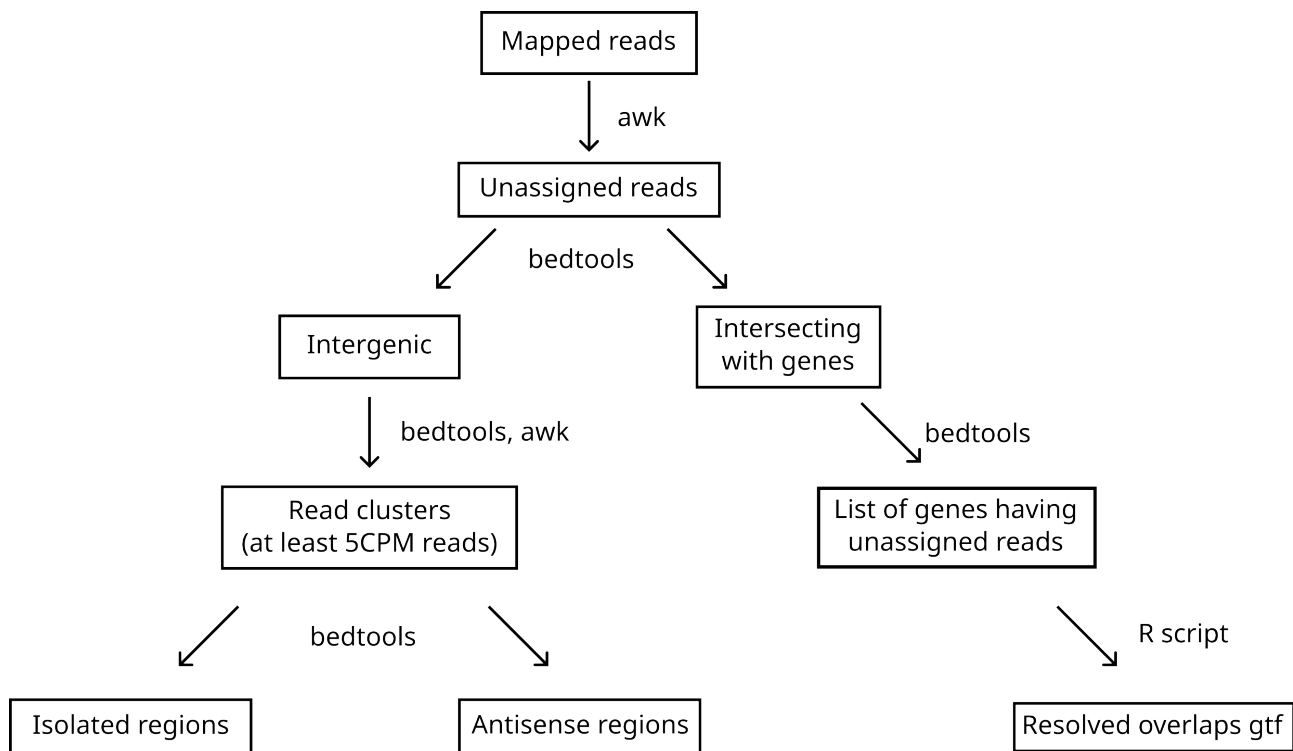


Figure 5.1: Scheme of the pipeline. This scheme provides only general logic and main tools of data processing. The pipeline itself is more complex, e.g., using multiple transcriptomic references required looping over some of those steps.

- ii. From the second reference and further: add genes to the original GTF that contain unassigned reads and do not overlap with entries from the original one.
- (b) For intergenic:
 - i. Cluster.
 - ii. Filter-out relatively small clusters (custom threshold).
 - iii. For the first reference only: filter-out AT-rich reads.
 - iv. For reads that are left, repeat from the beginning with the next reference.
 - v. For the last reference only: clusters that start just after 3' ends are assigned to genes (i.e., extend genes).
- 4. Create final GTF and map initial sequences to it.
- 5. Create list of large ($>5\text{CPM}$) intergenic clusters.

Mapping and filtering bam files Reads were mapped using STARsolo (Kaminow, Yunusov, and Dobin 2021), both from FASTQ and BAM input files. The same parameters were used for all samples (see [GitHub repository](#)), except the ones regarding barcode structures (differences include number of barcodes, lengths of barcode/UMI sequences, and their positions in the primer). Filtering unassigned reads was done using *awk*, taking those that have valid barcode (i.e. filtering out those reads that have barcode length not equal to the defined length) and were not assigned to any gene (i.e. had 'GN:Z:-' tag). Also, only uniquely mapped reads were taken (tag 'NH:i:1').

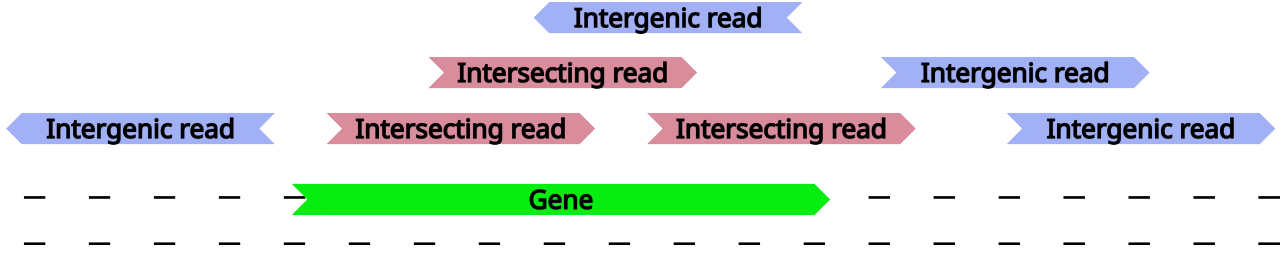


Figure 5.2: Unassigned read classification as intergenic or intersecting. Reads are defined as intersecting if they overlap with known genes and are on the same strand. Remaining reads are defined as intergenic.

Classifying reads as 'intergenic' or 'intersecting' and reads clustering Intersections of unassigned reads with references were checked using *bedtools intersect* command. Those that intersect with genes were classified as 'intersecting', and those that do not – as 'intergenic'. Intersections were assessed in strand-specific manner (*bedtools* '-s' flag), i.e. regions on different strands were treated as not intersecting, even if they share overlapping genomic coordinates (see Figure 5.2). The intergenic reads were clustered using *bedtools merge*, again in a strand-specific manner.

Manipulating transcriptomic references The overlapping gene resolving and construction of enhanced transcriptomic reference was done using custom R script, particularly *rtracklayer* library.

5.4 Intergenic regions

Extracting intergenic regions The intergenic reads were extracted as described in the previous section (i.e. taken unassigned reads that do not overlap with any reference used) and merged into clusters using *bedtools merge* command. Clusters were filtered based on cluster size, requiring at least 5 reads per million (CPM), where CPM represents the normalized read count per million of total primary reads in the dataset (i.e. $1CPM = 1 \times sample_size / 1000000$). The filtering was done using *awk*.

Cluster locations were adjusted using *deeptools* (to compute coverage) and custom python script, to avoid reads containing long introns making the intergenic clusters very wide. To accomplish this, for each intergenic region, the maximum coverage location was found and extended to include neighbouring regions that had at least half the maximum coverage. In such way, intergenic regions were adjusted to cover only 'peaks' of reads.

The lists acquired from all samples were then merged using *bedtools merge* function. This combined list then was filtered to contain only regions detected at least in all samples of one group ('eye', 'lung', 'brain', 'PBMC_10x' or 'PBMC_indrops'). Such filtering still allows to include tissue-specific regions in the combined list, while removing those that are not persistent across the same type of samples. Additionally, for each entry it was determined whether it overlaps with predicted genes from UCSC gene prediction archive (using *bedtools merge*) and distances to the closest genes were found (*bedtools closest*). Intergenic regions that were located on the opposite strand of known genes, were classified as *antisense*, while those that were not were classified as *isolated* (see Figure 5.3 for illustration).

This filtered combined list was then converted into GTF format and unassigned reads from each sample were mapped using this combined intergenic annotation.



Figure 5.3: Classification of intergenic regions as antisense or isolated.

Analysis of cell-gene matrices The matrices produced by STAR were filtered, allowing cells that have sufficient number of reads (thresholds selected manually), and only genes that were expressed in at least 3 cells. Also cells were filtered based on mitochondrial gene count (allowing up to 10% of mitochondrial gene counts in a cell). Doublets were filtered using *scrublet*. Afterwards, matrices were normalized (using *normalize_total* function from *scanpy* package) and log-transformed (i.e. for each entry $x = \log(x + 1)$, *log1p* function from *scanpy*).

Then only highly-variable genes ($\text{min_mean}=0.0125$, $\text{max_mean}=3$, $\text{min_disp}=0.5$) were selected. For the clustering, principal components (PCs) were computed and optimal number of them were selected based on elbow rule (manually). For blood samples, cells were annotated automatically using CellTypist (for other samples, annotation was skipped, as it was not in the main focus of this project). Then visualizations were made using UMAP embeddings (functions from the same *scanpy* package).

All the scripts with descriptions can be found in the [GitHub repository](#).

Correlations Spearman’s rank correlation between intergenic region and gene on the opposite strand was checked using custom python script. Formula for Spearman’s correlation is given by:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

where d_i is the difference between the ranks of corresponding values and n is the number of observations.

Oxford Nanopores data Nanopores data were mapped using *minimap2* algorithm and possible transcript structures were inferred using *stringtie* tool.

5.5 Enhancing transcriptomic reference

Three aspects were considered while enhancing transcriptomic reference:

- **Gene overlaps** – some of the genes are overlapping, resulting in unassigned reads from those overlapping regions.
- **3’ end extensions** – for some genes, there are clusters of reads just after their 3’ ends, and Pool et al. (2023) suggested to include those reads by extending 3’ ends for some genes.
- **Different annotations** – while most of the annotations agree on most genes, there is a small fraction of genes that are present in some references but not in others. Hence, for the enhancing transcriptomic reference, several transcriptomic references were used.

Overlaps The criteria for the resolving of overlapping genes are following:

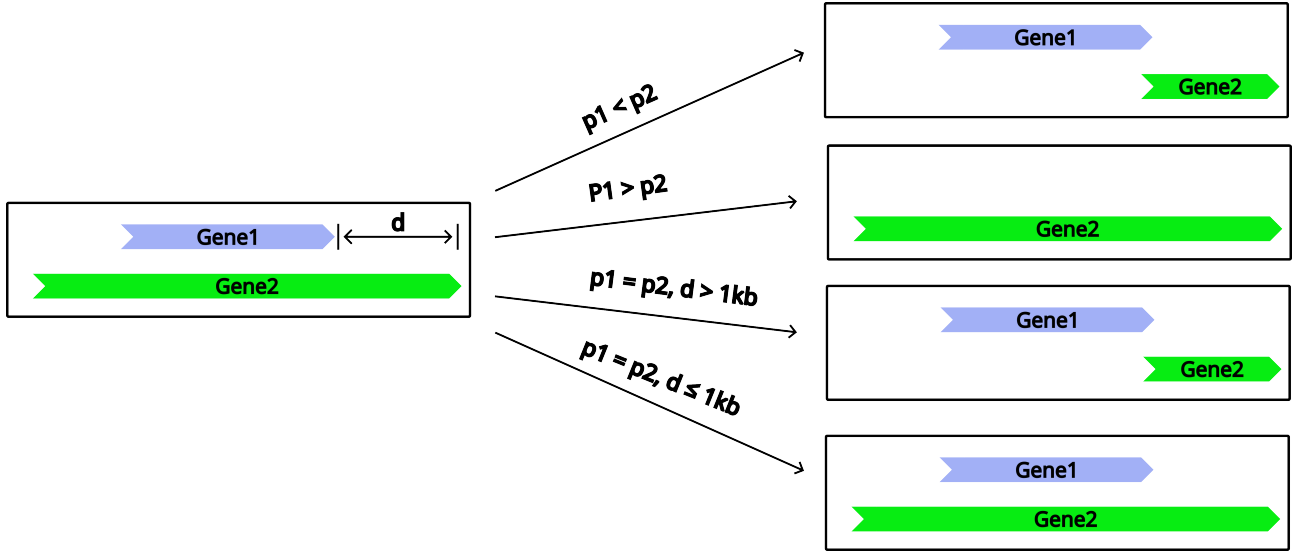


Figure 5.4: Strategy for resolving overlaps. $p1$ and $p2$ are priority scores of *gene1* and *gene2* (lower score means higher priority), while d is distance between 3' ends of those genes.

1. **Gene type:** prefer protein coding genes over other types, lncRNA over remaining (e.g. pseudogenes).
2. **Level:** some annotations have 'level' field, indicating if the gene is verified (score 1), manually annotated (score 2) or automatically annotated (score 3). Lower 'level' score was preferred.
3. **Intersection types:** if 5' end gene is overlapping with 3' end of gene, 5' end of gene was shortened.

Particularly, gene types were divided into three groups: protein coding, long non-coding RNAs and other, and each class was assigned score (see Table 5.3). The overall priority score was defined as sum of this gene type score and 'level' field. The lower this score is, the higher priority gene has. The intersection type (whether 3' end is overlapping or not) was taken into account only in cases when priority scores were equal, and distances between 3' ends of overlapping genes were greater than 1kb (see Figure 5.4 for graphical explanation). If possible, lower priority genes were shortened, otherwise they were deleted.

Class	Score	Gene Types
protein coding	0	protein_coding, IG_C_gene, IG_D_gene, etc.
long non-coding RNA	10	lncRNA, lincRNA
other	20	snRNA, snoRNA, pseudogene, etc.

Table 5.3: Priority scores assigned based on gene type.

3' end extensions Unassigned reads clusters (of size at least 1CPM) that were close to the 3' of genes (closer than 100bp) were assumed to originate from those genes. Hence annotations for those genes were expanded to cover those clusters.

Multiple annotations The initial transcriptomic reference (taken from 10x Genomics website) was enriched with entries from other annotations. Particularly entries from GENCODE, RefSeq and Inc-pedia transcriptomic references were added to the initial reference, if they overlapped with unassigned reads and did not overlap with genes already present in the reference. This enrichment was done sequentially, i.e. firstly the reference was updated with entries from GENCODE annotation, then resulting reference was updated with entries from RefSeq annotation and finally from Incpedia annotation.

5.6 Unmapped reads

Additionally, unmapped reads were extracted using samtools and mappings were searched against viral genome sequences (BLAST tool). The tool was run locally, with the viral genome sequences for database construction acquired from [NCBI Refseq genomes database](#).

6. RESULTS

6.1 Intergenic regions

Fourteen datasets from various tissues (blood, brain, eye, and lung) were analyzed. These datasets contained between 11% and 29% of uniquely mapped reads that remained unassigned. (the table with statistics can be found in Appendix 14.1) when using standard reference (taken from 10x Genomics). The majority of those unassigned reads were intergenic, meaning they did not overlap with any reference used. Given the dataset sizes, this proportion of reads could potentially contain biologically relevant information.

Analysing intergenic regions For each sample, intergenic regions containing a sufficient number of unassigned reads were identified, as described in the methods section (with a threshold set at 5 CPM). The term *intergenic* is used here in a strand-specific manner, meaning an intergenic region may be located on the opposite strand of a known gene.

To determine whether these intergenic regions contain biologically meaningful information, cells in each sample were clustered based solely on reads from these regions. As shown in selected examples in Figure 6.1, clustering was observed and roughly corresponded to clustering based on the standard (*10x Genomics*) annotation. This indicates that biologically meaningful information is indeed present in these intergenic regions. Consequently, further analyses were performed on these regions.

The lists of intergenic regions from all samples were combined and filtered based on the number of samples in which they were detected, resulting in a final set of 2,590 intergenic regions. Of these, 147 did not overlap known genes on the opposite strand and will be referred to as *isolated*. Those located on the opposite strand of known genes will be referred to as *antisense*.

A-rich sequences near the intergenic regions 3' end If there is a poly-A region near the 3' end of the intergenic region, then the presence of those peaks can be explained by mispriming, i.e., that would suggest that our intergenic region is not necessarily the actual 3' end of the novel transcript, but may be in the middle of it.

This is indeed the case for most of the reported intergenic regions, see Figure 6.2 for visual example and some statistics. The poly-A region here is defined as a region with at least 10 consecutive A's, allowing one mismatch.

Correlations Another aspect to assess is whether expression of those intergenic regions correlate with the expression of nearby genes. If so, this may be due to several factors:

- If the region correlates with gene on the same strand:

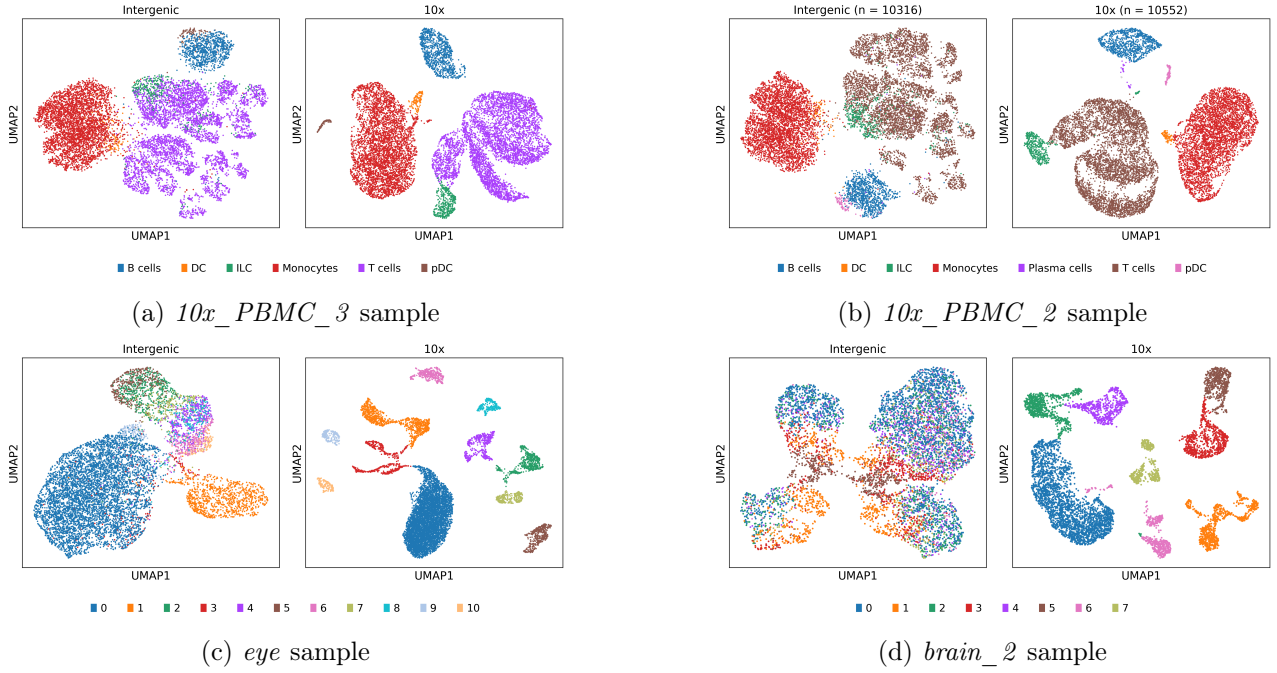


Figure 6.1: Comparison of clustering using standard annotation (*10x Genomics* reference) and using only defined intergenic regions. For some samples, intergenic regions are sufficient for rough clustering. For others (e.g., the *brain_2* sample), noise introduces clustering artifacts. Nevertheless, this rough clustering implies that these unassigned reads contain biological information.

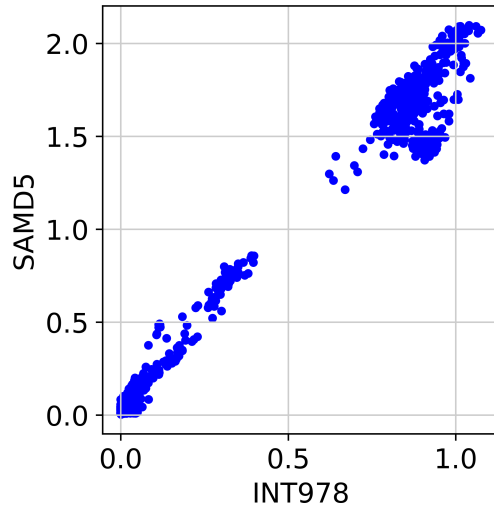
- Technical issues – the genome annotation might be not comprehensive and not include some longer transcripts that would span intergenic region as well as the gene with which correlation is observed.
- Biological reasons – genes in close proximity tend to be coexpressed due to the shared chromatin structure.
- If the region correlates with gene on the opposite strand:
 - Technical artifacts – errors during library preparation could produce inversed cDNA (will be discussed later).
 - Biological reasons – as mentioned before, genes in close proximity tend to be coexpressed.

It is difficult to determine which of these scenarios applies in our samples. Out of all defined intergenic regions, 97 (3.74 %) showed Spearman correlation >0.5 ($p < 0.05$) in at least 1 sample (83 correlated with genes on the opposite strand, 19 with a gene on the same strand, 5 with both). Some examples can be seen in Figure 6.3.

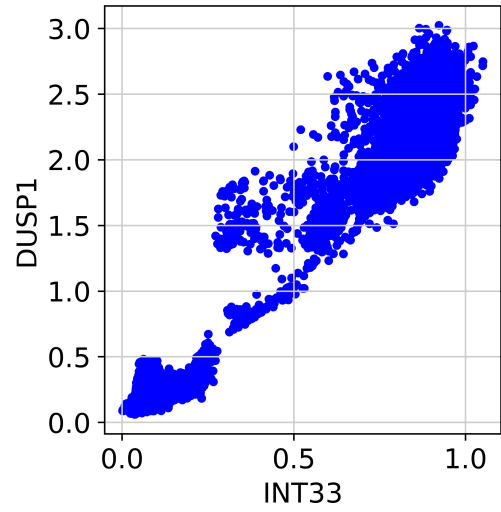
In the following subsections isolated and antisense intergenic regions will be discussed separately.

6.1.1 Isolated intergenic regions

The intergenic regions that are distant from known genes and contain reads may indicate novel genes. To determine whether these are genuine signals rather than artifacts, several aspects were considered:



(a) Correlation can be clearly seen in INT33 and DUSP1 normalized expressions (Spearman correlation 0.68) in the *PBMC_10x_2* sample, both of them are on the same strand.



(b) Correlation of INT978 and SAMD5 normalized expressions (Spearman correlation 0.85) in the *eye_3* sample, intergenic region is on the opposite strand of the gene.

Figure 6.3: Examples showing correlations between intergenic regions and nearby genes.

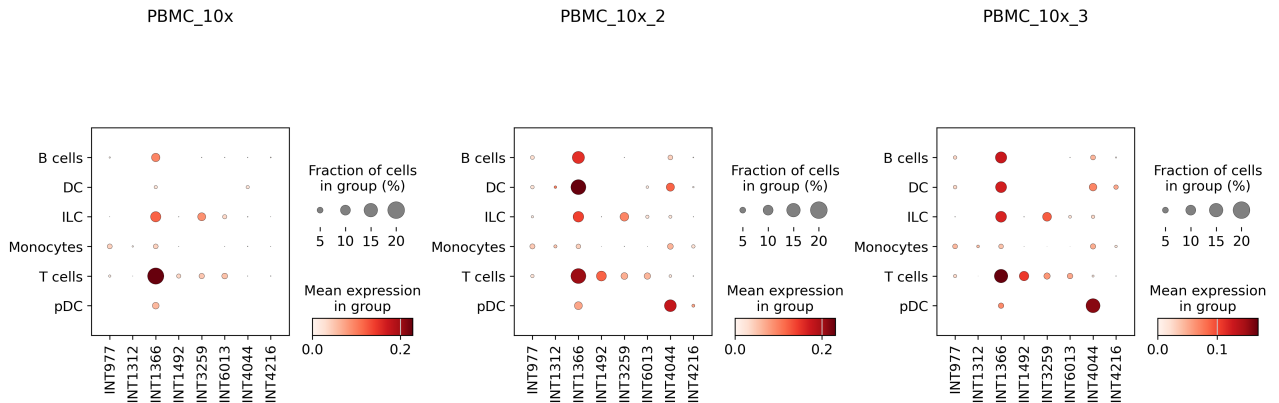


Figure 6.4: Examples of differentially expressed isolated intergenic regions in *PBMC_10x* samples.

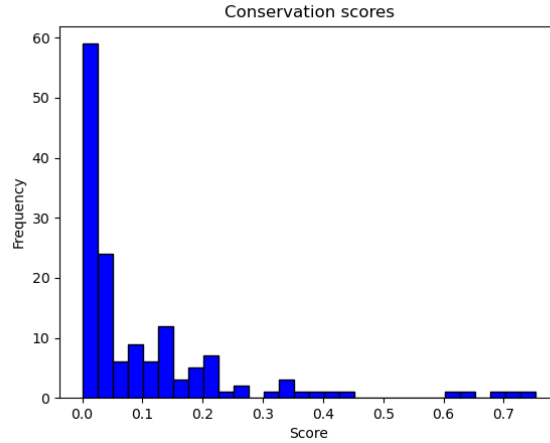


Figure 6.5: Histogram showing conservation scores in the isolated regions.

Open chromatin Open chromatin regions may provide further evidence of transcriptional activity when located upstream. However, distances to the nearest known genes must also be considered, as upstream open chromatin sites may not be associated with the intergenic regions but rather with other genes.

For each intergenic region, distances to the nearest open chromatin sites and upstream genes (strand-unspecific) were calculated. Of the 147 isolated intergenic regions, 66 had open chromatin regions closer than any known upstream genes. To ensure that these sites were not merely associated with distant genes, an additional filter was applied, requiring that the closest open chromatin region be at least 2,000 bp from the nearest known gene. This yielded 40 intergenic regions.

It should be noted that PBMC ATAC data was used for open chromatin regions, thus this feature should be taken with precaution for regions derived from other datasets.

Conservation scores The conservation score shows evolutionary conservation and is computed comparing genomic sequences from different species. A high conservation score in an intergenic region suggests potential functionality, providing additional evidence for biological relevance. To identify such regions, those with conservation scores above 0.6 were defined as conserved (there were 5 of such isolated regions). The overall conservation scores were quite low, suggesting that if the isolated intergenic genes originate from unannotated genes, they are probably human specific. The histogram showing conservation scores in isolated regions is given in the Figure 6.5.

Gene predictions To assess whether isolated intergenic regions overlap with predicted genes, UCSC gene prediction archives were examined, focusing on overlaps with 3' ends of predicted genes.

Of the identified regions, 12 overlapped with predicted genes. However, three of these did not overlap with the 3' ends of predicted genes, which would be expected given that the data was generated using 3' end sequencing methods. Five predicted genes were extensions of known genes, while the remaining four corresponded to genes absent from RefSeq and NCBI references but supported by our data. Table summarizing these findings can be found in Appendix 14.2.

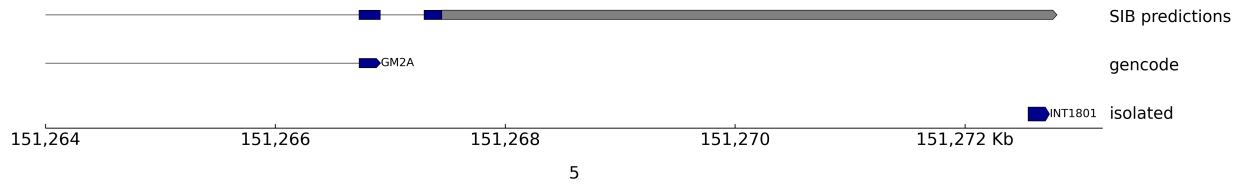


Figure 6.6: Example of gene which is predicted to have some longer transcripts than are present in the gencode and ncbi annotations. The intergenic region is located after the 3' end of gene GM2A, however, SIB prediction tool suggest that longer transcripts should be included in the annotation of this gene.

Combining supporting features In total, four features were examined as supportive evidence for the biological significance of these intergenic regions: conservation, open chromatin presence, differential expression, and gene predictions. Based on these, four sublists of isolated intergenic regions were generated:

- Conserved regions
- Regions associated with open chromatin
- Differentially expressed regions
- Regions overlapping predicted genes

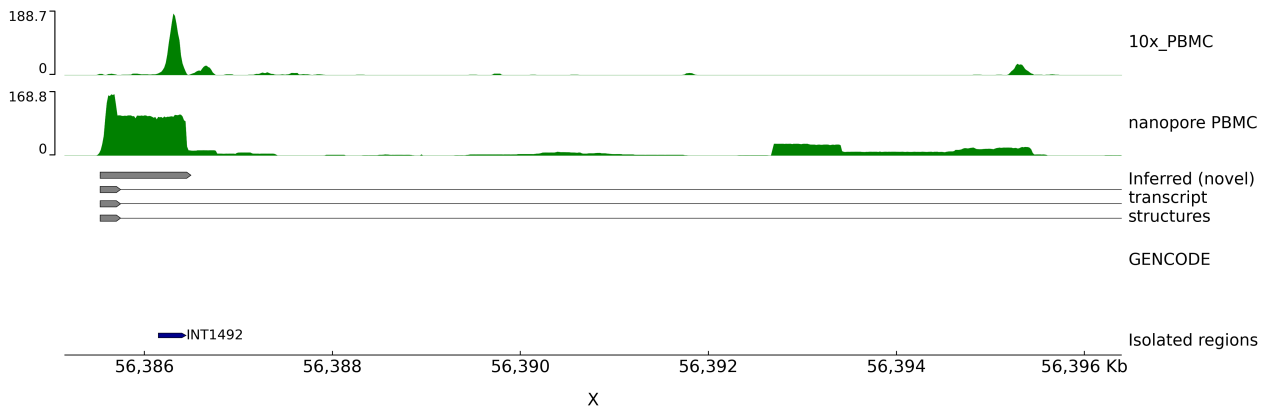
Notably, the list of conserved regions had only one element (INT1216) in common with the other lists (particularly, prediction list). The predicted gene list had two regions in common with the differential expression list (INT196, INT1525) and one in common with the open chromatin list (INT4147). All three of these predicted genes were extended versions of already annotated genes (see example in Figure 6.6). The differential expression and open chromatin lists had 11 regions in common (INT1312, INT1426, INT1492, INT1554, INT1609, INT1993, INT2209, INT349, INT368, INT4216, INT6013). No regions appeared in more than two lists.

For overall transcript structure inference, full-length Oxford Nanopore PBMC data were investigated. Isolated transcripts were also detected using this sequencing method, providing potential transcript structures. However, due to low coverage, these inferred structures should be interpreted with caution. An example of inferred transcript structures can be seen in Figure 6.7.

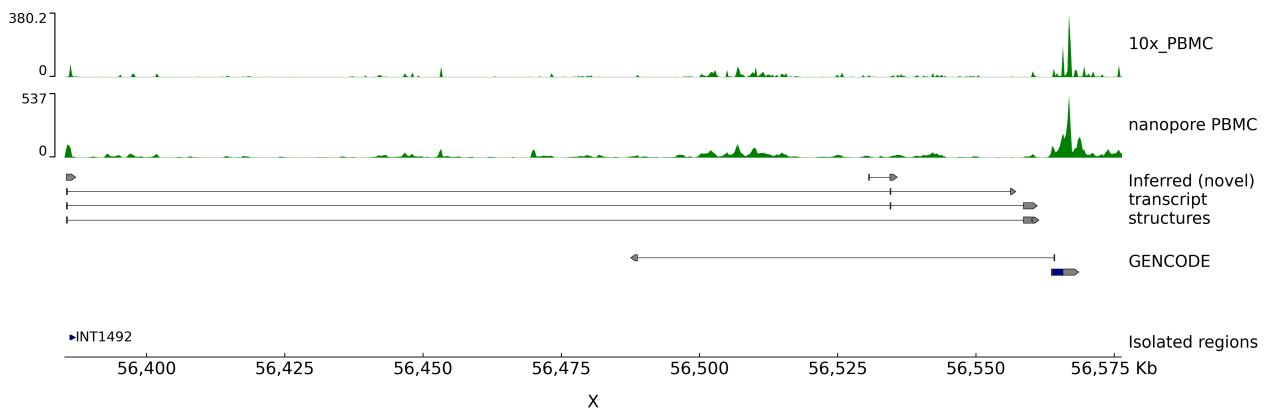
6.1.2 Antisense intergenic regions

Interpreting antisense intergenic regions is more complex. The features such as open chromatin or conservation are not strand specific, meaning that if they would be present, they could not be attributed with certainty for our intergenic regions. The differential expression can be checked, however, even if the antisense intergenic region is differentially expressed, it could still be an artifact, originating from differentially expressed transcripts. Hence from the features checked for the isolated intergenic regions, only real supporting evidence would be computational gene prediction.

Gene predictions As in the case of isolated intergenic regions, antisense intergenic regions were checked for overlaps with predicted genes from UCSC prediction archive. There were 49 such regions



(a) Inferred transcript structures of INT1492 region, zoomed-in view.



(b) Inferred transcript structures of INT1492 region, wider view.

Figure 6.7: From the Oxford Nanopore data several transcripts were inferred in the INT1492 region. The inferred transcripts varied in length, from around 1 kb to over 150 kb.

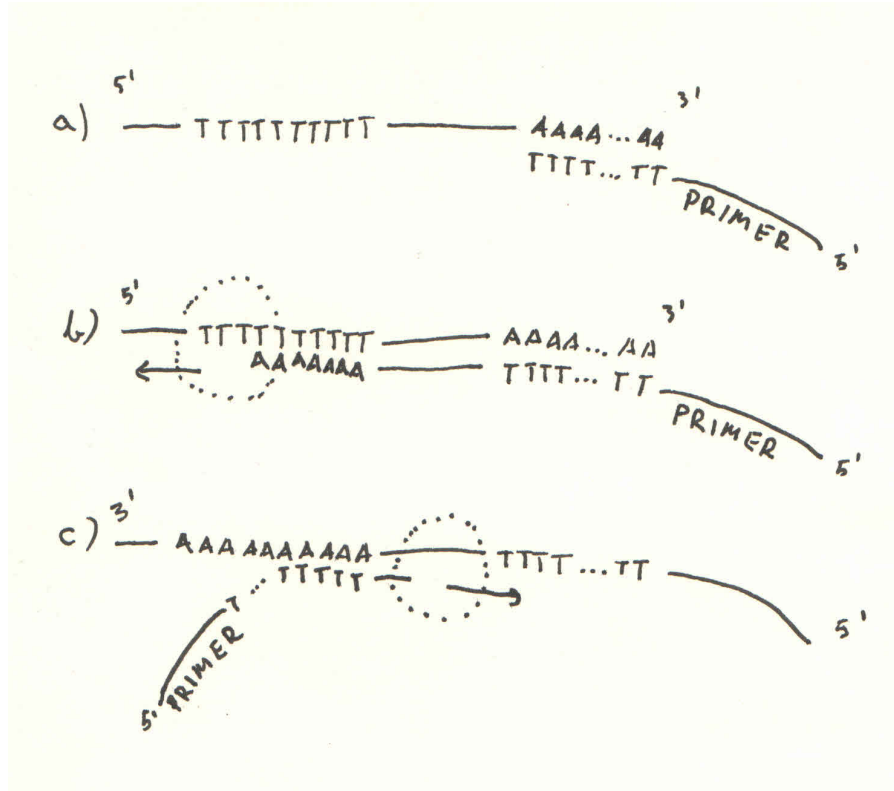


Figure 6.8: Scheme of poly-A mispriming resulting in reads mapped to the opposite strand compared to the original gene. a) RNA is captured by a poly-A tail with a primer containing a poly-T sequence. b) Reverse transcriptase attaches to the primer and synthesizes cDNA from the RNA template. c) If the RNA-cDNA complex becomes unstable and dissociates, another primer may bind to the poly-A stretches on the cDNA, if present. This would result in reversed cDNA, causing the reads to be mapped to the antisense strand.

found, out of them 35 were in the 3' ends of predicted genes. 9 of those 35 predicted genes were longer version of already annotated genes in RefSeq or GENCODE references. The complete list can be found in the Appendix [14.3](#).

Poly-A segments Having in mind that majority of the antisense intergenic regions have poly-A stretch after 3' end, and some of them strongly correlates with genes on the opposite strands, it is possible that at least some of them are artifacts from the library preparation. Particularly, it could happen that poly-T primer binds to poly-A stretch of cDNA produced by other primer, and in such way inversed transcripts are produced. These are later mapped to antisense strands of known genes (see Figure [6.8](#)).

6.2 Enhancing transcriptomic reference

Keeping in mind that unassigned reads are also present in gene locations, we attempted to resolve overlapping genes in the annotations so that reads from these regions could be included in the downstream analysis. For overlap resolution, a simple set of rules was applied, primarily favoring protein-coding genes over others and considering the level field in the transcriptomic annotations (more details can be found in Methods [5.5](#)).

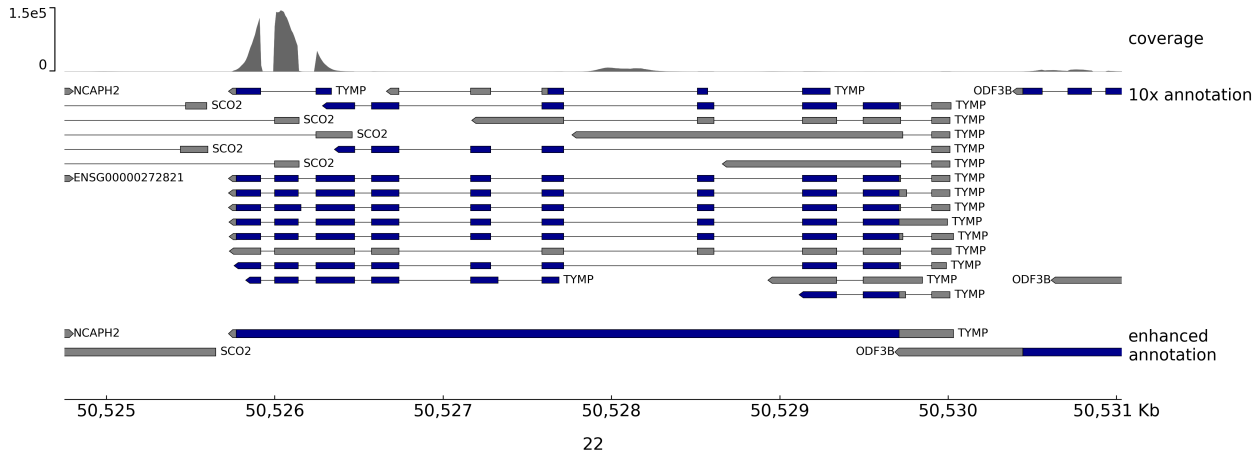


Figure 6.9: Example of successful resolution of overlapping genes TYMP and SCO2. Both genes are protein coding, however TYMP is preferred since it has lower *level* value. Based on the exon structures, it can clearly be seen that those reads in the overlapping area indeed originate from TYMP gene. Without reference enhancing, these reads from overlapping area would be classified as multimappers, and filtered out (in typical workflow). The 10x annotation track is expanded to highlight exon structures.

In addition to resolving overlaps, the initial transcriptomic reference (taken from 10x Genomics) was enriched with entries from other transcriptomic references (RefSeq, GENCODE, lncRNAPedia), ensuring that new entries did not overlap those already present. Finally, gene entries with reads near their 3' ends were extended.

This enhanced reference resulted in an average increase of 1.2% more reads in the cell-gene matrices. While this increase is relatively small, it enabled the detection of approximately 350 additional genes that were undetected using the standard transcriptomic reference (an example is given in Figure 6.9). However, an average of 13 genes per dataset were lost when using this enhanced reference.

Although the overall data volume increased, it is not straightforward to conclude that data quality improved. Overlap-handling rules can be debated, particularly regarding their applicability in all cases.

For example, genes are expressed in certain cell types or states regardless of their level classification (whether verified, manually annotated, or automatically annotated). While, in general, verified or manually annotated genes are more frequently expressed than automatically annotated ones (which are likely less explored and detected at lower rates overall), this pattern does not hold in all situations. The same consideration applies to gene classification by type.

Regarding the gene extensions, in most cases they increase the gene expression by relatively small fraction of counts (see Figure 6.10 for example). Even though some additional information is gained, when compared with the total amount of data, this gain is very small (e.g., in *PBMC_10x_2* sample, it constituted only 1.76% of gained reads, or 0.017% of total reads).

Overall, while statistical differences are evident, their impact on biological interpretations remains uncertain and likely depends on the specific downstream analysis. For instance, cell clustering is generally unaffected, and a visual inspection of UMAPs does not reveal significant differences between datasets generated using the enhanced and standard references (see example in Figure 6.11). However, in certain cases, even a small number of additional genes could significantly influence the analysis, meaning that this enhanced reference could help uncover relevant biological processes.

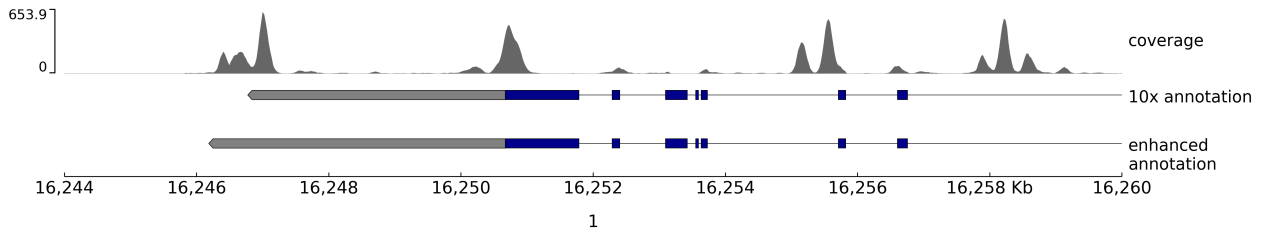


Figure 6.10: Example of gene extension (coverage shown is from *PBMC_10x_2* dataset). In this case gene FBX042 is extended to include reads close to its 3' end. However, the amount of reads gained is tiny when compared with the number of total counts of this gene (the number of counts of this gene increased approximately by 3% in the *PBMC_10x_2* dataset).

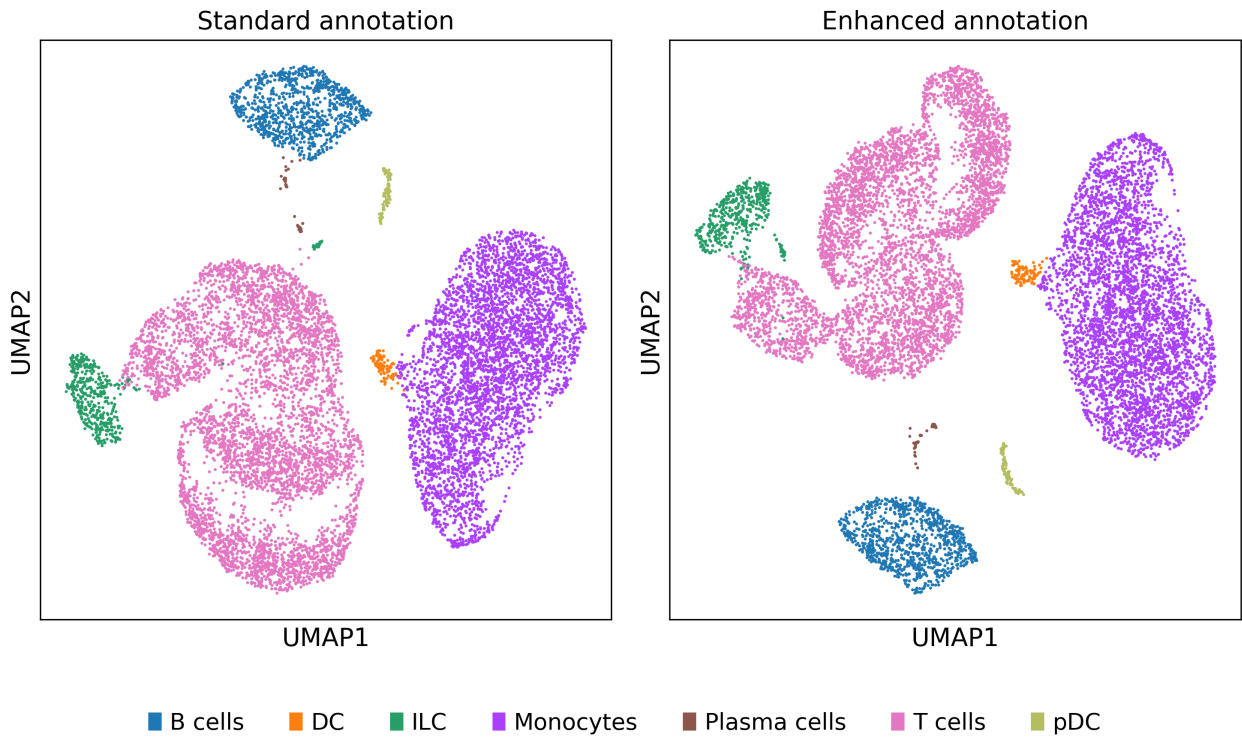


Figure 6.11: UMAPs of *PBMC_10x_2* dataset using standard and enhanced references. While T cell cluster is a bit different in these two cases, main structures remain the same in both cases.

6.3 Unmapped reads

In the previous sections, mapped but unassigned reads were explored. However, in scRNA-seq data, there are also reads that do not map to the reference genome at all. In the analyzed datasets, such reads constituted approximately 3–13% (mean 5.96%) of total reads. The majority of these are likely technical artifacts. Nonetheless, it is worth investigating whether some may represent meaningful sequences originating from non-human DNA, such as viral genomes.

To explore this, unmapped reads were aligned to viral genomes using the BLAST tool. Indeed, a small fraction (around 1–2% of unmapped reads) mapped to viral genomes. The viral species to which reads mapped were consistent across samples, with most reads aligning to *BeAn 58058 virus*, *Human endogenous retrovirus K113*, and *Aeribacillus phage AP45*.

While it is difficult to draw definitive conclusions from this type of mapping, these findings suggest that scRNA-seq data is sufficiently sensitive to detect non-host transcripts even in healthy individuals.

7. DISCUSSION

Single-cell RNA sequencing is a widely used tool for analyzing cellular systems nowadays. These experiments generate vast amounts of data, and since they are becoming increasingly affordable and accessible, the data is often used solely for the primary goal of the study, without deeper exploration. One underexamined aspect in most studies is the presence of reads that map to the genome but are not assigned to any gene. As shown in the Results chapter, such reads can comprise 10–30% of total reads—a significant portion of the dataset.

This project aimed to investigate these unassigned reads. They were first classified based on whether they intersected with known genes or not (referred to as intergenic). Intergenic reads were then divided into two groups: antisense (located on the opposite strand of known genes) and isolated (with no known genes on either strand). Reads were clustered, and only regions with at least 5 CPM were analyzed.

Antisense intergenic regions Two key observations — expression correlations with genes on the opposite strand and poly-A stretches downstream of the regions — led us to hypothesize that these reads may be artifacts produced during the reverse transcription step. These reads represent the vast majority of unassigned reads, so addressing this mechanism could improve the efficiency of single-cell RNA-seq experiments. It should be noted that this hypothesis applies specifically to 3'-end capturing methods, which rely on poly-A tails.

Temperature adjustments might help reduce such artifacts: at lower temperatures, RNA-cDNA complexes are more stable. In the 10x Genomics v3.1 protocol, reverse transcription is performed at 52°C, while in inDrops2 it occurs at 42°C, where we observed fewer unassigned reads (see Appendix 14.1). However, strong conclusions cannot be drawn, as only two samples from inDrops2 were analyzed.

While this hypothesis may explain the majority of antisense reads, other possibilities should not be dismissed. For instance, template switching — when reverse transcriptase (or polymerase) detaches and reattaches elsewhere — is reported in both reverse transcription and amplification steps (Houseley and Tollervey 2010; Potapov and Ong 2017). Although artifacts from template switching are unlikely to form large clusters (due to expected more random distribution), they may explain reads in clusters smaller than 5 CPM, which were not explored in this project.

Isolated intergenic regions Unlike antisense regions, isolated regions might represent unannotated (novel) genes rather than artifacts. This is particularly likely for regions with supporting features, the most indicative of which is differential expression — since noise is not expected to be cell-type specific.

Although the human genome has been extensively studied, several factors still complicate novel gene identification. First, since protein-coding genes have more defined structures (and are thus easier

to detect computationally) (Guigó 2023), these potential novel genes are likely non-coding, such as lncRNAs. Second, their low expression and high cell-type specificity hinder confident annotation, which is generally required by widely used transcriptome references like GENCODE or RefSeq.

While transcript structures can be inferred using long-read RNA sequencing (as we demonstrated using Oxford Nanopore data), high sequencing depth is needed for confident annotations.

Although lncRNAs can participate in virtually any cellular process, predicting the function of a specific lncRNA is challenging. They often lack clear phenotypic effects, and mutations typically influence quantitative traits (Mattick et al. 2023). This project did not attempt to assign functions to these novel candidates; future studies could explore this.

A subset of isolated regions located near known genes could be explained by unannotated transcript extensions. Some overlapped predicted longer isoforms of known genes, suggesting this as a possible origin. Long-read sequencing data can be used to validate this hypothesis.

Enhancing transcriptomic reference Some reads were unassigned despite aligning to annotated gene regions. This is mainly due to gene overlaps. To incorporate such reads, the transcriptomic reference was enhanced by resolving overlaps based on gene types. Using this enhanced reference, downstream cell-gene matrices increased by $\sim 1.2\%$ on average.

While overall assigned reads and gene detection increased, some genes showed reduced counts after overlap resolution — highlighting the complexity of the transcriptome and the difficulty of disambiguating gene overlaps with short reads.

Our method for enhancing transcriptomic references raises several points of consideration: first, prioritizing gene types (e.g., protein-coding over lncRNAs) can be subjective, and may not always be appropriate. Second, our overlap resolution used only gene-level annotations (start and end coordinates), whereas transcriptome annotations typically include exon-intron structures and isoforms.

Alternative strategies for handling multi-mapped reads exist — such as uniform distribution or probabilistic assignment based on uniquely mapped read ratios (Deschamps-Francoeur, Simoneau, and Scott 2020). However, no consensus or gold standard currently exists, so such reads are often excluded to reduce uncertainty.

An improved approach could consider transcript-level annotations, e.g., comparing read coverage with exon structures to infer the most likely gene of origin in overlapping regions. Such a tool could recover more reads and improve assignment accuracy.

Unmapped reads As a side investigation, unmapped reads were also briefly explored. While the majority are likely artifacts, some may originate from non-human organisms, such as viruses. We found that a small fraction aligned to viral species — suggesting that even unmapped reads may hold biological value and should not be dismissed outright.

Limitations and discrepancies This project analyzed datasets generated using short-read, 3' end-capturing RNA-seq methods. Datasets from other methods would exhibit different artifacts and require separate analysis. These data did not allow the definition of full transcript structures, which is possible with full-length sequencing.

An interesting discrepancy is shown in Table 14.1: the proportion of unassigned reads is lower when computed from raw counts compared to demultiplexed counts. Since demultiplexed counts

represent actual molecule counts, and raw counts reflect amplification, one would expect similar ratios — assuming equal amplification efficiency.

However, we observed a consistently higher proportion of unassigned reads in the demultiplexed data, suggesting that unassigned reads may be amplified less efficiently. The origin of this bias is unclear and warrants further study. This effect is especially pronounced in datasets generated using inDrops2, so comparisons between inDrops2 and 10x v3.1 may help uncover its cause.

8. CONCLUSIONS

- Single cell RNA sequencing data contain reads that map to the genome, but are not assigned to any gene. In the analyzed datasets, these unassigned reads accounted for 11–29% of all reads.
- The unassigned reads mapping to the antisense strand of known genes constitute the majority of unassigned reads and are likely artifacts produced during the reverse transcription reaction. This observation is specific to 3' end sequencing protocols.
- Unassigned reads mapping to regions without annotated genes on either strand may originate from unannotated genes. This thesis proposes 147 such candidate regions. These candidate regions should be validated through other type of experiments.
- Some unassigned reads mapped to regions with overlapping genes may be recovered by resolving gene overlaps in the transcriptomic annotation. The proposed approach increased the number of counts in the cell–gene matrices by approximately 1.2%.
- Overall, the results suggest that at least part of unassigned reads might be biologically meaningful and should not be discarded by default.

9. DISCLAIMER

Parts of this thesis were reviewed using OpenAI's ChatGPT to assist with grammar and style corrections. The scientific content, interpretations, and conclusions are solely my own.

10. ACKNOWLEDGEMENTS

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12. SUMMARY

Vilnius University
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ENHANCING SINGLE-CELL RNA-SEQ ANALYSIS THROUGH THE INTEGRATION OF UNACCOUNTED INTERGENIC REGIONS

Juozapas Ivanauskas

Abstract

Single-cell RNA sequencing (scRNA-seq) experiments generate large datasets, a portion of which often consists of reads that align to the genome but are not assigned to any annotated gene. This project aimed to investigate the origins of these unassigned reads and explore strategies to recover them for downstream analysis.

Unassigned reads from fourteen publicly available scRNA-seq datasets were analyzed. The majority were found to originate from the antisense strand of known genes. It is hypothesized that these reads are artifacts introduced during the reverse transcription step, particularly in 3'-end capturing protocols. Another subset of reads mapped to intergenic regions without any annotated genes on either strand. These regions may correspond to unannotated transcripts, and 147 such candidate regions were identified for future validation.

Additionally, a fraction of unassigned reads mapped within known gene regions but were not assigned due to overlapping annotations. To recover these, the transcriptomic reference was enhanced by resolving gene overlaps, extending 3' ends of selected genes, and combining information from four different annotations. This modification led to an average increase of 1.2% in total read counts in the resulting cell-gene matrices.

Together, these findings suggest that unassigned reads—often discarded in scRNA-seq workflows—can be informative and should be systematically examined to improve transcriptomic references and the interpretability of single-cell data.

13. SUMMARY IN LITHUANIAN

Vilniaus Universitetas
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PAVIENIŲ LĄSTELIŲ RNR SEKOSKAITOS DUOMENŲ ANALIZĖS TOBULINIMAS INTEGRUOJANT NEAPRAŠYTUS TARPGENINIUS REGIONUS

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Santrauka

Pavienių ląstelių RNR sekoskaitos eksperimentai sugeneruoja didelius duomenų kiekius, kurių dalį sudaro sekos, priligiuojamos prie genomo, bet nepriskiriamos jokiame genui. Šio projekto tikslas buvo ištirti šias sekas ir pabandyti įtraukti jas į tolimesnes analizes.

Šiame projekte buvo išanalizuota nepriskirtos sekos iš keturiolikos duomenų rinkinių. Dauguma jų buvo priligiuojamos prie priešingos DNR grandies nei žinomi genai. Buvo iškelta hipotezė, kad šios sekos yra artefaktai susidarantys atvirkštinės transkripcijos metu (ši hipotezė galioja tik 3' galo sekoskaitos metodams). Kita dalis sekų buvo priligiuota prie genomo vietose, kuriose iš viso nėra žinomų genų. Šios sekos galimai kyla iš neaprašytų genų. 147 regionai buvo iškelti kaip kandidatai naujiems genams, kuriuos reikėtų patvirtinti papildomais eksperimentais.

Galiausiai, dalis sekų buvo priligiuojamos tose vietose, kur yra žinomų genų, tačiau vis tiek liko nepriskirtos. Kad įtrauktume šias sekas į tolimesnę analizę, patobulinome transkriptominį anotacijos failą. Tai buvo pasiekta išsprendžiant persidengiančius genus, pratęsiant kai kurių genų 3' galus bei naudojant informaciją iš kelių skirtingų transkriptomo anotacijų. Taikant šį metodą, duomenų kiekis tolesnėje analizėje padidėjo vidutiniškai 1.2%.

Šie rezultatai rodo, kad nepriskiriamos sekos (įprastai neįtraukiamos į tolimesnes analizes), gali būti informatyvios ir turėtų būti sistemaiškai išanalizuotos siekiant pagerinti dabartinės genomo anotacijas bei pavienių ląstelių duomenų analizę.

14. APPENDICES

Sample	Total Reads	Unassigned	Intergenic	Demultiplexed	Unassigned	Intergenic
PBMC_10x	182330834	15.95%	11.37%	70305137	18.51%	12.86%
PBMC_10x_2	496387931	18.52%	14.26%	195948926	21.29%	16.32%
PBMC_10x_3	368640939	18.59%	14.34%	167804193	20.99%	16.17%
PBMC_indrops	112932507	11.12%	6.5%	8176578	22.86%	10.25%
PBMC_indrops_2	471705924	14.87%	8.36%	98063027	25.52%	13.73%
brain	206360627	16.87%	10.67%	143916177	19.02%	12.07%
brain_2	122556503	22.32%	15.98%	95031864	23.77%	17.00%
eye	375397270	28.02%	18.97%	161882445	31.87%	21.60%
eye_2	140981808	29.35%	14.85%	69146995	31.43%	15.99%
eye_3	161261977	29.46%	18.86%	68157696	32.11%	20.58%
lung_2	511080104	15.99%	12.07%	269118747	17.98%	13.33%
lung_5	452105505	14.49%	10.74%	256353198	16.19%	11.83%
lung_7	524095146	18.41%	14.34%	236157103	20.64%	15.79%
lung_8	342092138	14.53%	10.66%	217801055	16.07%	11.67%

Table 14.1: Statistics of unassigned and intergenic reads per sample. Unassigned reads were counted after mapping with 10x reference. Intergenic reads here are those that do not intersect with any reference used. 'Demultiplexed' column shows number of total reads after demultiplexing using UMIs, and following 'unassigned' and 'intergenic' percentages were computed compared to this number.

Name	Genomic coordinates	Prediction tool	Overlaps with 3' region of predicted gene
INT196	6:89086200-89086500	SIB	Yes (PNRC1)
INT827	9:94111950-94112200	SIB	Yes
INT1216	1:72282700-72282900	SIB	No (5' end)
INT1387	11:63570800-63571050	SIB	No
INT1525	8:11842200-11842350	SIB	Yes (CTSB)
INT1801	5:151272550-151272700	SIB	Yes (GM2A)
INT2044	19:4041150-4041400	SIB	Yes (ZBTB7A)
INT4070	4:47430500-47430700	SIB	Yes
INT4147	1:34861500-34861700	SIB	Yes (DLGAP3)
INT4577	5:702050-702300	SIB	Yes
INT4948	5:703250-703450	SIB	Yes
INT5710	4:47428500-47428700	SIB	No

Table 14.2: Isolated intergenic regions overlapping with predicted genes from UCSC gene prediction archive. The gene in the brackets shows if the predicted gene is extended version of already annotated genes.

Table 14.3: Antisense intergenic regions overlapping with predicted genes from UCSC gene prediction archive. The gene in the brackets shows if the predicted gene is extended version of already annotated genes.

Name	Genomic coordinates	Prediction tool	Overlaps with 3' region of predicted gene
INT7	6:24718850-24719100	SIB	Yes
INT33	5:172767250-172767500	SIB	Yes (DUSP1)
INT134	15:41280550-41280800	SIB	Yes (extends a bit after)
INT218	5:61409150-61409400	SPG	No (5' end)
INT285	10:110898200-110898550	SIB	Yes (BBIP1)
INT327	1:169132100-169132350	SIB	Yes (NME7)
INT347	14:75277700-75278000	Gscan	No
INT382	12:11892250-11892500	Gscan	Yes
INT386	6:26215200-26215450	SIB	Yes (H2BCB?)
INT438	4:39713300-39713600	SIB	Yes (almost all (short) gene spanned)
INT533	8:130110900-130111150	SIB	Yes
INT617	15:85733950-85734200	SIB	Yes
INT621	9:75148850-75149400	SIB	Yes (OSTF1)
INT651	1:28579100-28579350	SIB	Yes (TRNAU1AP)
INT654	6:34382850-34383050	SIB	No (but short gene)
INT828	8:133035800-133036050	SIB	Yes (SLA)
INT859	2:71373500-71373750	SIB	No (slight overlap on 5' end)
INT898	17:51190400-51190700	SIB	Yes (extends after)
INT903	8:22613400-22613650	SIB	Yes
INT949	9:111693050-111693400	SIB	Yes
INT972	15:64954800-64955350	SIB	Yes
INT984	18:63291900-63292200	Geneid	Yes
INT1028	5:50414200-50414500	SIB	Yes
INT1155	1:36296050-36296300	SIB	No (5' end)
INT1231	10:19890700-19891050	Gscan	Yes
INT1402	5:142786700-142786950	SIB	No (5' end)
INT1440	2:235746450-235746700	SIB	Yes
INT1445	9:40885650-40885900	SIB	Yes
INT1548	7:50342700-50342950	Gscan	No (exon in the middle)
INT1616	17:30895450-30895700	SIB	No (5' end)
INT1749	3:172333550-172333750	SIB	No (5' end)
INT1931	17:82522100-82522350	SIB	Yes
INT2065	10:103606150-103606400	SIB	Yes
INT2158	14:49636100-49636350	SIB	No (5' end, DNAAF2)
INT2194	7:130521250-130521500	SIB	No (but gene is short)
INT2371	1:14632650-14632900	Geneid	No (middle exon)
INT2380	22:33853900-33854150	Geneid	Yes
INT2825	13:45300050-45300300	Geneid	No (middle exon)
INT2979	4:98929350-98929600	SIB	No (5' end, EIF4E)
INT3328	2:184604250-184604500	SIB	Yes
INT3429	17:48172750-48173150	Augustus	Yes
INT3504	15:25332950-25333150	SIB	Yes (UBE3A)
INT3849	3:121432200-121432400	SIB	Yes
INT3868	17:31448500-31448700	SIB	Yes
INT4458	5:44820700-44820900	SIB	Yes (MRP530)
INT4743	20:3188900-3189050	SIB	Yes (DDRKG1)
INT5002	11:92232750-92232950	SIB	Yes
INT5315	20:3183750-3184050	SIB	Yes
INT5329	13:95603000-95603200	SIB	Yes (extends after)